Class 10 NCERT Maths

CLASS 10 NCERT MATHS

Chapter 4: Quadratic Equations

CHAPTER 4: QUADRATIC EQUATIONS

A simplified, detailed guide in Hinglish
For students who missed Class 7-8-9
Based on the official NCERT English Medium textbook

SECTION 1: FOUNDATION

1.1 POLYNOMIAL REVISION (Chapter 2 se)

Yaad hai? Polynomial = variable + power = whole no.
- Linear: $ax + b$ (degree 1)
- QUADRATIC: $ax^2 + bx + c$ (degree 2) <- ye chapter
- Cubic: $ax^3 + \ldots$ (degree 3)

1.2 EQUATION VS EXPRESSION

- Expression: $2x^2 + 3x - 5$ (sirf likha hua)
- Equation: $2x^2 + 3x - 5 = 0$ ('=' sign hai)

1.3 QUADRATIC EQUATION KYA HAI?

DEFINITION: Wo equation jiska standard form:

$$ax^2 + bx + c = 0$$

jahan a, b, c real numbers AND $a \neq 0$.

IMPORTANT: $a \neq 0$ warna linear ban jayegi.

QUADRATIC EXAMPLES:

- $x^2 - 5x + 6 = 0$ ($a=1, b=-5, c=6$)
- $2x^2 + 7x = 0$ ($a=2, b=7, c=0$)
- $3x^2 - 12 = 0$ ($a=3, b=0, c=-12$)

NOT QUADRATIC:

- $2x + 5 = 0$ (no x^2)
- $x^3 + 2x + 1 = 0$ (degree 3)
- $x^2 + 1/x = 5$ ($1/x = x^{-1}$, negative power)

1.4 ROOTS OF A QUADRATIC EQUATION

ROOT = value of x jiske liye LHS = RHS = 0.

Quadratic equation ke 2 roots hote hain (max).

EXAMPLE: $x^2 - 5x + 6 = 0$
Check $x = 2$: $4 - 10 + 6 = 0$ (ok)
Check $x = 3$: $9 - 15 + 6 = 0$ (ok)

So roots are 2 and 3.

SECTION 2: FACTORISATION METHOD

2.1 IDEA

Quadratic ko $(x - p)(x - q) = 0$ form mein todo.
Phir:
Either $x - p = 0$ -> $x = p$
Or $x - q = 0$ -> $x = q$

LOGIC: Agar do cheezon ka product 0 hai, toh
kam-se-kam ek toh 0 hi hogi.

2.2 SPLITTING MIDDLE TERM

$ax^2 + bx + c = 0$ mein:
Middle coefficient b ko aise 2 nos. p, q mein todo
ki:
$p + q = b$ AND
$p \times q = a \times c$

EXAMPLE 1: $x^2 + 7x + 12 = 0$

$a=1$, $b=7$, $c=12$, so $a \times c = 12$.
Find p, q with $p + q = 7$, $p \times q = 12$.

Try: 3 and 4 -> $3+4=7$, $3 \times 4=12$ (ok)

$x^2 + 7x + 12 = 0$
$x^2 + 3x + 4x + 12 = 0$
$x(x + 3) + 4(x + 3) = 0$

$(x + 3)(x + 4) = 0$

So $x = -3$ or $x = -4$.

EXAMPLE 2: $2x^2 - 5x + 3 = 0$

$a=2$, $b=-5$, $c=3$, $a \times c = 6$.
Find p, q with $p + q = -5$, $p \times q = 6$.

Try: -2 and -3 -> -5, 6 (ok)

$2x^2 - 2x - 3x + 3 = 0$
$2x(x - 1) - 3(x - 1) = 0$
$(x - 1)(2x - 3) = 0$

So $x = 1$ or $x = 3/2$.

EXAMPLE 3: $6x^2 - x - 2 = 0$

$a=6$, $b=-1$, $c=-2$, $a \times c = -12$.
$p + q = -1$, $p \times q = -12$.

Try: -4 and 3 -> -1, -12 (ok)

$6x^2 - 4x + 3x - 2 = 0$
$2x(3x - 2) + 1(3x - 2) = 0$
$(3x - 2)(2x + 1) = 0$

So $x = 2/3$ or $x = -1/2$.

SECTION 3: QUADRATIC FORMULA

3.1 THE MAGIC FORMULA (YAAD RAKH)

For $ax^2 + bx + c = 0$:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Ye formula HAR quadratic pe kaam karta hai.

3.2 DISCRIMINANT D KA KAMAAL

$D = b^2 - 4ac$

D ki value se roots ka nature pata chalta hai:

D ki value	Roots
$D > 0$	Two DISTINCT REAL roots
$D = 0$	Two EQUAL real roots (repeated)
$D < 0$	NO REAL roots (imaginary)

3.3 EXAMPLES

EXAMPLE 4: $2x^2 - 7x + 3 = 0$

a=2, b=-7, c=3
D = 49 - 24 = 25 (>0, distinct real)

  x = (7 +/- sqrt(25)) / 4
  x = (7 +/- 5) / 4

So x = 3 or x = 1/2.

EXAMPLE 5: $x^2 - 6x + 9 = 0$

a=1, b=-6, c=9
D = 36 - 36 = 0 (equal roots)

  x = (6 +/- 0) / 2 = 3

So x = 3 (repeated).

EXAMPLE 6: $x^2 + x + 1 = 0$

a=1, b=1, c=1
D = 1 - 4 = -3 (<0)

So NO real roots.

SECTION 4: WORD PROBLEMS

EXAMPLE 7: Number problem

"Ek number aur uska reciprocal ka sum 10/3 hai.
Number nikaalo."

Let number = x.
  x + 1/x = 10/3

Multiply by 3x:

$3x^2 + 3 = 10x$

$3x^2 - 10x + 3 = 0$

Splitting (a x c = 9, sum = -10):

p = -9, q = -1

$3x^2 - 9x - x + 3 = 0$

$3x(x - 3) - 1(x - 3) = 0$

$(x - 3)(3x - 1) = 0$

So x = 3 or x = 1/3.

(Dono valid - reciprocals of each other)

EXAMPLE 8: Speed problem

"Ek train 360 km fixed speed se cover karti hai.
Agar speed 5 km/h zyada hoti, toh time 1 hour
kam lagta. Original speed?"

Let original speed = x km/h.

Original time = 360/x

New time = 360/(x+5)

Equation:

$360/x - 360/(x+5) = 1$

Multiply by x(x+5):

$360(x+5) - 360x = x(x+5)$

$1800 = x^2 + 5x$

$x^2 + 5x - 1800 = 0$

Splitting (a x c = -1800, sum = 5):

45 x (-40) = -1800, sum = 5 (ok)

$(x + 45)(x - 40) = 0$

x = -45 (rejected, speed +ve)

x = 40 km/h (Answer)

Q AND A TIME

Q1. Inn mein se konsi QUADRATIC equation hai?

(a) $x^2 + 5 = 0$

(b) $2x + 3 = 0$

(c) $x^3 - 1 = 0$

(d) $5x^2 - 7x + 1 = 0$

Q2. $x^2 - 3x - 10 = 0$ ko factorisation se solve kar.

Q3. $2x^2 + x - 6 = 0$ ko quadratic formula se solve kar.

Q4. Discriminant nikaal aur roots ka nature bata:
(a) $x^2 + 4x + 4 = 0$
(b) $2x^2 - 3x + 5 = 0$
(c) $x^2 - 5x + 4 = 0$

Q5. Word problem: "Do consecutive odd positive integers ka product 143 hai. Numbers nikaalo."

SUMMARY

---

1. Quadratic equation: $ax^2 + bx + c = 0$ ($a \neq 0$)

2. Roots: Maximum 2 hote hain

3. Methods:
   - FACTORISATION (splitting middle term)
   - QUADRATIC FORMULA: $x = (-b \pm \sqrt{D}) / 2a$

4. Discriminant $D = b^2 - 4ac$:
   - $D > 0$ -> distinct real roots
   - $D = 0$ -> equal roots
   - $D < 0$ -> no real roots

5. Word problems:
   Variable le -> equation banao -> solve -> impossible solutions reject karo.
